



Crypto Analytics Dashboard

A full-stack MERN application that provides real-time cryptocurrency analytics, portfolio tracking, and market insights. The platform helps users monitor crypto market trends, analyze historical data, and manage their personal watchlists.



Problem Statement

Cryptocurrency markets are highly volatile and generate massive amounts of data every second. Investors often need to switch between multiple platforms to track prices, market capitalization, trading volumes, and portfolio performance.

The goal of this project is to build a centralized Crypto Analytics Dashboard that enables users to:

- Monitor real-time cryptocurrency prices.
- Analyze market trends using interactive charts.
- Track top gainers and losers.
- Manage a personalized watchlist.
- Monitor portfolio performance.
- Access detailed information about individual cryptocurrencies.

This project is designed for learning advanced MERN Stack concepts, API integration, authentication, caching, and data visualization.



Objectives

- Build a production-ready MERN application.
 - Learn third-party API integration.
 - Implement authentication and authorization.
 - Create reusable dashboard components.
 - Visualize data using charts and graphs.
 - Practice backend optimization and caching.
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Features



Authentication & Authorization

- User Registration
- User Login
- JWT Authentication
- Protected Routes
- Profile Management



Market Dashboard

- Global Crypto Market Overview
- Total Market Capitalization
- Total Trading Volume
- BTC Dominance
- Market Trend Indicators



Coin Search

- Search cryptocurrencies by name
- Search by symbol
- Instant search suggestions



Coin Details

- Coin Name
- Symbol
- Rank
- Current Price
- Market Capitalization
- Trading Volume
- Circulating Supply
- All-Time High (ATH)
- All-Time Low (ATL)



Analytics & Charts

- Price History Charts
- Market Cap Trends
- Trading Volume Analysis
- Historical Performance



Market Insights

- Top Gainers
- Top Losers
- Trending Coins
- Most Viewed Coins



Watchlist

- Add Coins to Watchlist
- Remove Coins from Watchlist
- Personalized Dashboard



Portfolio Tracker

- Add Holdings
- Portfolio Value Calculation
- Profit/Loss Tracking
- Performance Analytics

UI Features

- Responsive Design
- Dark Mode
- Light Mode
- Modern Dashboard Layout

Performance Features

- API Response Caching
 - Pagination
 - Lazy Loading
 - Optimized Database Queries
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Tech Stack

Frontend

- React.js
- Redux Toolkit
- React Router
- Axios
- Tailwind CSS
- Chart.js / Recharts

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT Authentication
- Redis (Optional)

APIs

- CoinGecko API
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Project Structure

```
crypto-dashboard/  
|  
├─ client/  
|   └─ src/  
|       ├─ components/  
|       ├─ pages/  
|       ├─ redux/  
|       └─ services/
```

```
|   |   └─ utils/
|
|   └─ server/
|       └─ controllers/
|           └─ routes/
|               └─ middleware/
|                   └─ models/
|                       └─ services/
|                           └─ config/
|
└─ README.md
```

Database Collections

Users

```
{
  name,
  email,
  password,
  watchlist,
  portfolio
}
```

Watchlist

```
{
  userId,
  coinId,
  addedAt
}
```

Portfolio

```
{
  userId,
  coinId,
  quantity,
  purchasePrice
}
```

API Endpoints

Authentication

```
POST /api/auth/register
POST /api/auth/login
GET /api/auth/profile
```

Coins

```
GET /api/coins
GET /api/coins/:id
GET /api/coins/trending
GET /api/coins/gainers
GET /api/coins/losers
```

Watchlist

```
POST /api/watchlist
GET /api/watchlist
DELETE /api/watchlist/:coinId
```

Portfolio

```
POST /api/portfolio
GET /api/portfolio
PUT /api/portfolio/:id
DELETE /api/portfolio/:id
```

Learning Outcomes

By building this project, you will learn:

- MERN Stack Architecture
- REST API Development
- JWT Authentication
- MongoDB Data Modeling
- Redux Toolkit State Management
- API Integration
- Chart Visualization
- Caching Strategies
- Performance Optimization
- Responsive UI Design
- Production-Level Folder Structure

Future Enhancements

- Real-time WebSocket Updates
 - AI-Based Market Prediction
 - Crypto News Feed
 - Multi-Currency Support
 - Email Notifications
 - Mobile Application
 - Social Trading Features
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Project Level

Difficulty: Advanced

This project is ideal for developers who want hands-on experience building a real-world analytics dashboard using the MERN stack.